

Joon-Hyeok Yim

☎ Contact: +82-10-4172-0629
✉ Email: joonhyeokym@gmail.com

Position

2025 – 2026 **Yale University**, New Haven, CT, United States
Postdoctoral Associate, Department of Electrical & Computer Engineering

Education

2016 – 2025 **Yale University**, New Haven, CT, United States
Doctor of Philosophy in Mathematics, Advisor: *Prof. Anna C. Gilbert*
Dissertation Title: On the Analysis of Average Hyperbolicity and Its Applications
Personal Leave (2018 – 2021): for mandatory military service (Technical Research Personnel)

2012 – 2016 **Seoul National University**, Seoul, Republic of Korea
Bachelor of Science in Mathematical Sciences, *Summa Cum Laude* (3.97/4.3)

Work Experience

2018 – 2025 **Teaching Fellow**, *Yale University*, New Haven, CT, United States.
Instructor: Calculus of Functions of Several Variables (Fall 2022)
Teaching Assistant: Discrete Mathematics (Fall 2024, Spring 2025), Analysis I (Spring 2024), Linear Algebra with Applications (Spring 2022), Linear Algebra and Matrix Theory (Spring 2018)
Coach: Calculus of Functions of One Variable II (Spring 2023)

2018 – 2021 **Technical Research Personnel**, Republic of Korea.
Alternative Military Service: R&D at private-sector companies, focusing on NLP and CV

Papers

Fitting trees to ℓ_1 -hyperbolic distances (with Anna C. Gilbert). Advances in Neural Information Processing Systems 36 (NeurIPS 2023)

Hyperbolicity, slimness, and minsize, on average (with Anna C. Gilbert). preprint arXiv:2412.05746

A quantitative tree-likeness bound from average hyperbolicity. preprint arXiv:2609.18023

Towards stratified sampling for redistricting plans (with Zijian Wang, Gregory J. Herschlag, Jonathan C. Mattingly, and Anna C. Gilbert). Submitted arXiv:2609.18741

Active Projects

Probabilistic methods for structured discrete models.

Evaluation and data for AI mathematical reasoning.

Property testing for metric and combinatorial structures.

Honors & Awards

2012 – 2016 **Presidential Science Scholarship**, *Korea Student Aid Foundation*, Republic of Korea.

2013, 2014, 2015 **Undergraduate Mathematical Competition in Korea**, *Gold Prize*.
Republic of Korea

2010, 2011 **International Mathematical Olympiad**, *Silver Medal*.

Selected Talks & Sessions

- Sep 2026 **From δ -hyperbolicity to tree approximation**, *DGIST*, Daegu, Republic of Korea, Theoretical Computer Science Seminar
- Jan 2025 **Slim and fat triangles in random graphs**, *Seoul National University*, Seoul, Republic of Korea, Probability Seminar
- Mar 2024 **Tree fitting problem and hyperbolic distances**, *Yonsei University*, Seoul, Republic of Korea, Discrete Analysis Seminar
- Dec 2023 **Fitting trees to ℓ_1 -hyperbolic distances**, *NeurIPS 2023*, New Orleans, LA, United States, Poster Session
- Oct 2023 **Fitting trees to ℓ_1 -hyperbolic distances**, *Yale FDS/Google Workshop: Theory and Practice of Foundation Models*, New Haven, CT, United States, Poster Session

Conferences & Workshops

- Feb 2025 **Fusing Theory and Practice of Graph Algorithms**, *ICERM*, Providence, RI, United States
- Dec 2023 **NeurIPS 2023**, New Orleans, LA, United States
- Oct 2023 **Yale FDS/Google Workshop: Theory and Practice of Foundation Models**, *Yale University*, New Haven, CT, United States
- Jun 2023 **Modern Applied and Computational Analysis**, *ICERM*, Providence, RI, United States
- Dec 2021 **Geometric and Topological Methods in Data Science**, *ICERM*, Providence, RI, United States

Skills

- Programming Python
- Language Korean (native), English (fluent)